IOT-BASED SMART GRID SYSTEM USING ESP32 FOR REAL TIME MONITERING AND CONTROL

A PROJECT REPORT

Submitted by

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In partial fulfillment for the award of the degree

Of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

GOVERNMENT COLLEGE OF ENGINEERING

(AUTONOMOUS)

BARGUR, KRISHNAGIRI – 635 104

NOVEMBER -2025

GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMUS)

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ACKNOWLEDGEMENT

We feel glad to take this opportunity to cordially acknowledge a number of people who provided us a great support during our project.

We would like to express our deep sense of gratitude to our respected Principal Dr.V.Thirunavukkarasu,M.E.,Ph.D., who has bestowed his kind grace and affection of us in accomplishing this project.

Sincere thanks to Dr.V.Thirunavukkarasu,M.E.,Ph.D Head of the Department- Electronics and Communication Engineering in Government College of Engineering Bargur, on providing facility to do the project work and utilize all the facilities in this college.

We would like to thank our Project Co-ordinator Mrs.NARGIS M.E - Electronics and Communication Engineering, who has been a great mentor and gave the spirit to us to continuously focus the project.

We make use of this opportunity to express gratitude and sincere thanks to our project guide Mr.,V.RAJESHKANNAN M.E who directed us throughout the entire phase of our project with her esteemed presence and knowledge. It's her motivation and guidance which made us to explore and gain knowledge in our project.

Finally we thank all the teaching and non-teaching staffs and laboratory assistants for their constructive guidance and encouragement.

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ABSTRACT

The IoT-Based Smart Grid System using ESP32 is designed to monitor and control electrical power usage efficiently through Internet of Things technology. The system measures voltage and current using sensors, calculates power using the formula P = V × I, and displays the data on an LCD or OLED screen. It employs LEDs to indicate different power levels — green for low, yellow for medium, and red for high consumption.

A relay module is used to automatically control the connected loads when the power exceeds a set threshold, ensuring safety and preventing overload conditions.The ESP32 microcontroller acts as the central unit, enabling real-time monitoring and communication through its in-built Wi-Fi module.

Power usage data can be viewed and controlled remotely using a mobile application such as Blynk, providing users with convenient access to their energy information. This project offers a low-cost, reliable, and scalable approach to smart energy management, promoting efficient power utilization and contributing to the development of modern smart grid systems for sustainable living.

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CHAPTER 1

1. INTRODUCTION

The increasing demand for efficient energy management and monitoring systems has led to the development of smart grid technologies that integrate sensing, control, and automation. In modern electrical systems, understanding real-time power consumption and optimizing load distribution are essential for reducing energy wastage and improving reliability. This project focuses on building an IoT-based smart monitoring system powered by a Li-ion battery, which measures voltage, current, and power using sensors and intelligently controls multiple loads through a relay module.

By utilizing a microcontroller such as the ESP32, the system can process sensor data, make automated decisions, and provide users with live information about energy usage.The proposed system connects various household or small-scale electrical loads such as a 9 W bulb, an electric fan, and an LED strip to demonstrate practical power control and monitoring. Voltage and current sensors continuously measure the electrical parameters, while the computed power values determine when to switch loads on or off based on predefined thresholds. This setup not only ensures efficient power utilization but also demonstrates how renewable or battery-based energy systems can be integrated into smart environments. Overall, the project aims to promote energy efficiency, automation, and awareness of power consumption in real-time.

2. EXISTING SYSTEM

In existing power monitoring and control systems, most setups rely on manual operation and conventional meters for measuring energy usage. Traditional electrical circuits typically use analog meters to indicate voltage and current, which provide only instantaneous readings without data logging or automatic control. These systems lack the ability to monitor real-time variations in power consumption or to manage loads automatically based on the available power supply.

As a result, users are unable to optimize energy usage effectively, leading to wastage and reduced battery backup time when operated on stored power sources like Li-ion batteries.Some existing automated systems use microcontrollers or programmable logic controllers (PLCs) but are often limited by high cost, complex wiring, or lack of wireless monitoring. Additionally, many of these setups are designed only for grid-connected systems and do not support portable or battery-powered applications. They usually require manual switching of loads and do not provide integrated power measurement with load control. Hence, there is a need for a low-cost, battery-powered, microcontroller-based system that can both monitor and control loads intelligently, making energy management more efficient and suitable for smart grid and IoT applications.

3. PROBLEM STATEMENT

In conventional electrical systems, there is no provision for real-time monitoring or automatic control of power consumption, leading to inefficient energy usage and increased power losses. Users often rely on manual switching of loads without knowing the actual voltage, current, or power being consumed. This lack of automation not only reduces the efficiency of energy utilization but also affects the lifespan of the power source, especially when using battery-based systems such as Li-ion packs.

Furthermore, traditional systems do not provide feedback or alert mechanisms to prevent overload conditions or optimize the distribution of available power among multiple loads.Therefore, there is a pressing need for a smart, automated system capable of measuring voltage, current, and power in real time, while controlling multiple loads efficiently based on the power availability and consumption levels. The system should be compact, reliable, cost-effective, and suitable for both grid-independent and battery-powered applications. Such a solution would help in better load management, reduce manual intervention, and enhance overall energy efficiency in modern smart grid environments

4. Proposed Solution

The proposed system is an ESP32-based smart power monitoring and control unit powered by a Li-ion battery. It continuously measures voltage, current, and power consumption using sensors such as ACS712 and a voltage divider circuit. A 4-channel relay module is used to control multiple electrical loads like a bulb, fan, and LED strip. Based on the measured power and predefined thresholds, the microcontroller automatically switches loads ON or OFF to prevent overloading and ensure efficient energy utilization.

Motivation

In today’s world, energy conservation and efficient power management have become essential due to the increasing demand for electricity and limited energy resources. Many households and small systems still rely on manual switching and conventional monitoring methods, which often lead to energy wastage and system inefficiency. The motivation behind this project is to design an automated and intelligent system that can monitor power consumption in real time and control loads efficiently without human intervention.By utilizing modern technologies like the ESP32 microcontroller, current and voltage sensors, and relay modules, the project aims to make energy usage more transparent and manageable. Implementing such a system not only promotes energy efficiency but also helps extend battery life in backup systems like Li-ion packs.

1.6.BACKGROUND OF THE STUDY

With the rapid advancement of technology and the growing dependence on electrical devices, the need for efficient energy monitoring and management systems has become increasingly important. Traditional electrical setups often rely on manual operation and conventional meters, which provide only basic readings without real-time analysis or automatic control. As a result, users are unable to track power consumption accurately or manage loads effectively, leading to unnecessary energy wastage and reduced system reliability.

To overcome these limitations, the integration of microcontrollers, sensors, and IoT technology has opened new possibilities for smart energy systems. Microcontrollers like the ESP32 allow real-time data processing and wireless communication, while sensors such as ACS712 and voltage dividers enable precise measurement of electrical parameters. By combining these technologies with automation through relay modules, it becomes possible to develop an intelligent, battery-powered system that monitors voltage, current, and power, and automatically controls loads for optimal energy utilization. This study focuses on designing such a system to demonstrate the practical application of smart grid concepts in small-scale environments.

CHAPTER 2

2.1. ESP 32 MICROCONTROLLER

The ESP32 is a powerful and versatile microcontroller developed by Espressif Systems, widely used in Internet of Things (IoT) applications due to its high performance, low power

consumption, and integrated wireless communication features. It serves as the brain of the IoT-enabled safety helmet, controlling all sensors and communication modules.

The ESP32 is equipped with a dual-core processor (Tensilica Xtensa LX6) that operates at up to 240 MHz, providing fast and efficient processing capabilities. It has built-in Wi-Fi (802.11 b/g/n) and Bluetooth (v4.2 + BLE) connectivity, enabling seamless wireless data transmission between the helmet and the monitoring station. This feature makes it ideal for real-time monitoring applications in industrial environments.

Key Features of ESP32:

Processor: Dual-core 32-bit Xtensa LX6 microprocessor (up to 240 MHz)

Memory: 520 KB SRAM and external flash memory support Connectivity:

Integrated Wi-Fi and Bluetooth/BLE

GPIO Pins: Multiple General Purpose Input/Output pins for interfacing sensors and modules

ADC/DAC: 12-bit ADC and 2-channel 8-bit DAC for analog signal processing

Power Efficiency: Supports deep sleep and low power modes

Operating Voltage: 3.0V – 3.6V Role

in IoT-Enabled Safety Helmet:

In the smart safety helmet, the ESP32 acts as the central processing unit. It collects

2.2 VOLTAGE SENSOR ZMTB101B

The ZMPT101B is a high-precision voltage sensor module designed to measure AC voltage accurately. It is based on the ZMPT101B voltage transformer, which provides electrical isolation between the high-voltage AC circuit and the low-voltage microcontroller system, ensuring user safety.

The module converts the AC voltage signal into a proportional analog voltage output that can be read by an ADC pin of microcontrollers such as the ESP32 or Arduino.This sensor is highly sensitive and suitable for measuring voltages up to around 250V AC, depending on the calibration. It includes an onboard potentiometer (variable resistor) that allows you to adjust the sensitivity or calibration to match your system's voltage range. The ZMPT101B module operates typically on 5V, though it can also work with 3.3V logic controllers with proper scaling.

2.3 CURRENT SENSOR ACS712

The ACS712 is a Hall-effect-based current sensor module used to measure both AC and DC currents accurately. It operates on the principle of the Hall effect, where a magnetic field produced by the current flowing through a conductor is detected by a Hall sensor. The sensor then converts this magnetic field into a proportional voltage output that can be read by the ADC pin of a microcontroller such as the ESP32 or Arduino.

The ACS712 module is available in three variants—5A, 20A, and 30A—depending on the current range required. It provides an analog output voltage that varies linearly with the measured current, with a sensitivity of about 185 mV/A (5A module), 100 mV/A (20A module), or 66 mV/A (30A module). The sensor operates at 5V and produces an output of 2.5V at zero current, with the output voltage increasing or decreasing based on the direction of current flow.

2.4 BATTERY

FIGURE2.5 BATTERY(LI-I0N)

The 18650 Li-ion battery is used as the main power source for the IoT-enabled safety helmet. It provides a stable voltage and high energy capacity, ensuring long operation time for all components like the ESP32, sensors, GPS module, and buzzer. This rechargeable battery is lightweight, reliable, and ideal for portable IoT devices.

The *Lithium-ion (Li-ion) battery* is the primary power source used in this project, chosen for its high energy density, lightweight design, and long cycle life. It provides a *stable DC voltage* to power the ESP32 controller, sensors, and relay module efficiently. The project uses a *3-cell (3S) Li-ion pack, which offers a nominal voltage of **11.1V* and can be charged up to *12.6V. A **Battery Management System (BMS)* is integrated with the pack to ensure safe operation by protecting against overcharging, over-discharging, and short circuits. The Li-ion battery offers a high discharge rate, making it suitable for running multiple loads such as a bulb, fan, and LED strip simultaneously.Compared to other rechargeable batteries like NiMH or lead-acid types, Li-ion batteries are more efficient, compact, and have a lower self-discharge rate. They deliver consistent performance even under variable load conditions, making them ideal for portable IoT and embedded systems. In this project, the battery output is passed through a *DC-DC converter* to regulate voltage levels suitable for the ESP32 (5V or 3.3V). This setup not only ensures stable power delivery but also enhances energy efficiency, allowing longer operation during off-grid or backup modes. Thus, the Li-ion battery serves as a reliable and sustainable power unit for the smart load monitoring and control system.

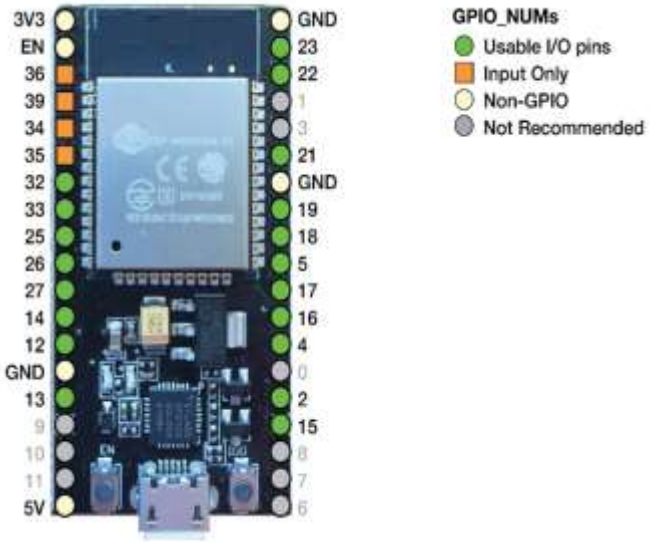
2.6 4-CHANNEL RELAY MODULE

A relay is an electromechanical switch used to control high-voltage or high-current circuits using a low-power signal from a microcontroller or control circuit. It consists of an electromagnet, armature, spring, and a set of contacts. When current flows through the coil, it generates a magnetic field that moves the armature, opening or closing the contacts to switch the connected load. This allows devices like Arduino or ESP32 to operate electrical appliances safely without direct contact with high power. Relays are widely used in automation, safety systems, and embedded applications for isolation, control, and automatic operation of electrical devices.

2.7 LCD DISPLAY

An LCD (Liquid Crystal Display) is an electronic display module used to show information such as text, numbers, and symbols in embedded systems. It works by controlling the light passing through liquid crystal cells using an electric field, allowing characters to appear on the screen. The most common type is the 16x2 LCD, which can display 16 characters per line on two lines. LCDs are energy-efficient, lightweight, and easy to interface with microcontrollers like Arduino or ESP32. In smart safety systems, the LCD display is used to present real-time sensor readings, alerts, and status messages to help users monitor conditions effectively.

ESP 32 PIN DIAGRAM



The **ESP32** is a powerful microcontroller that includes both **Wi-Fi** and **Bluetooth** connectivity. It has around **30 to 38 pins**, depending on the module type. These pins serve multiple purposes:

GPIO Pins (General Purpose Input/Output): Used to connect sensors, LEDs, and other components.

ADC (Analog to Digital Converter): Converts analog signals (like from sensors) to digital values.

DAC (Digital to Analog Converter): Converts digital data into analog signals.

PWM (Pulse Width Modulation): Used for controlling motors, LEDs, and brightness.

Communication Pins:

UART: For serial communication.

SPI: For high-speed data exchange with sensors and displays.

I2C: For connecting multiple modules using only two wires.

Power Pins: Includes 3.3V, 5V, and GND for powering external devices.

EN (Enable) Pin: Used to reset the ESP32.

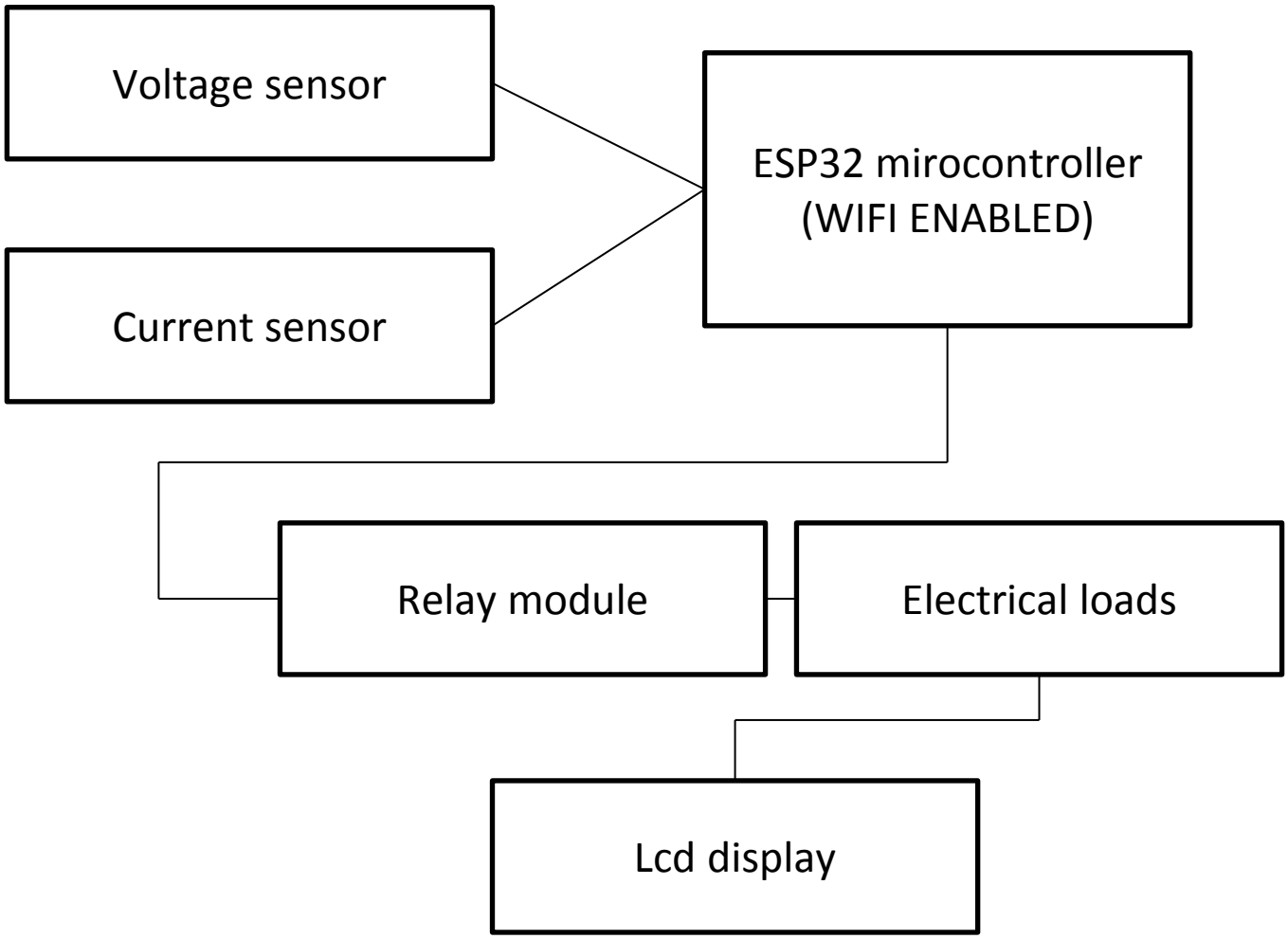
The pin configuration makes the ESP32 ideal for IoT and embedded projects requiring wireless communication and sensor integration.

SENSOR PIN CONNECTION

•all components are interconnected to the ESP32 microcontroller for effective monitoring and control. The voltage sensor (ZMPT101B) is connected with its VCC pin to the 3.3V/5V supply, GND to the common ground, and the output (OUT) pin to GPIO34 of the ESP32, which acts as an analog input to measure the AC voltage level. Similarly, the current sensor (ACS712) is powered by the 5V pin of the ESP32, with GND connected to the common ground, and its output pin connected to GPIO35 to read the analog current values. The 4-channel relay module is interfaced with the ESP32 to control different loads — IN1, IN2, IN3, and IN4 are connected to GPIO25, GPIO26, GPIO27, and GPIO14, respectively. The relay module's VCC is connected to 5V, and GND is tied to the ESP32 ground. The Li-ion battery pack provides the main power supply through a Battery Management System (BMS), ensuring protection and stable

•· output to the ESP32 and connected modules. This setup allows accurate voltage and current measurement, safe load control, and reliable operation of the smart grid system.

BLOCK DIAGRAM



Explanation:
Sensors (voltage ,curent): reads the curent and voltage consuption
ESP32: Processes sensor signals and decides actions.
Relay Module: Activates alarms, fans, or sprinklers in case of danger.
LCD Display: Shows real-time sensor readings and alerts for monitoring.

SYSTEM OPERATION

1. Core Components

Embedded System: A microcontroller (e.g., ESP32, Arduino) acts as the brain, executing the safety logic and coordinating all functions.

Sensors: Dedicated modules like the Voltage sensor (ZMTB101B), Current Sensor (ACS712).

Actuators: Local alarming devices like LEDs are used for immediate and visual alerts.

Connectivity: A Wi-Fi (IoT) or GSM/GPRS module enables remote communication.

2. Operational Cycle:

<u>Sensing:</u>

The voltage sensor (ZMPT101B) measures the supply or load voltage, and the current sensor (ACS712) measures the current drawn by each load.These analog signals are sent to the ESP32's ADC pins, which convert them into digital values for further analysis.

<u>Processing:</u>

The ESP32 microcontroller processes the sensor data and calculates real-time power consumption using $P=V\times I$P=V×I.It compares the measured power and battery voltage against predefined thresholds to decide load control actions.

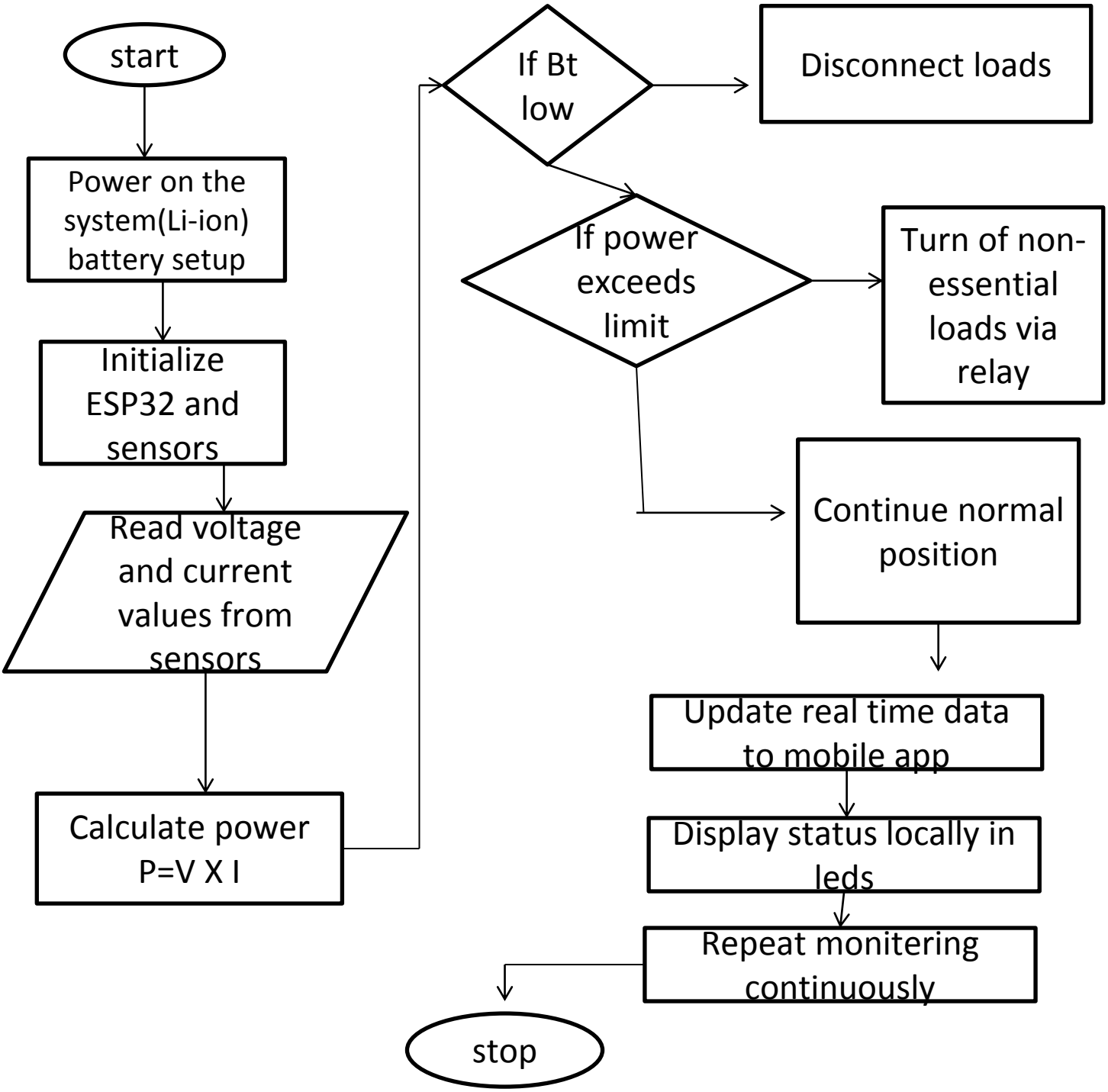
<u>Local Alert:</u>

When abnormal conditions such as overload or low battery occur, the system triggers visual or audible alerts using LEDs or a buzzer.The relay module automatically disconnects non-essential loads to prevent damage or excessive battery drain.

<u>Remote Notification:</u>

The ESP32 transmits the voltage, current, and power data to a mobile application via Wi-Fi (Blynk/MQTT).Users receive real-time updates and notifications, allowing remote monitoring and control of connected loads

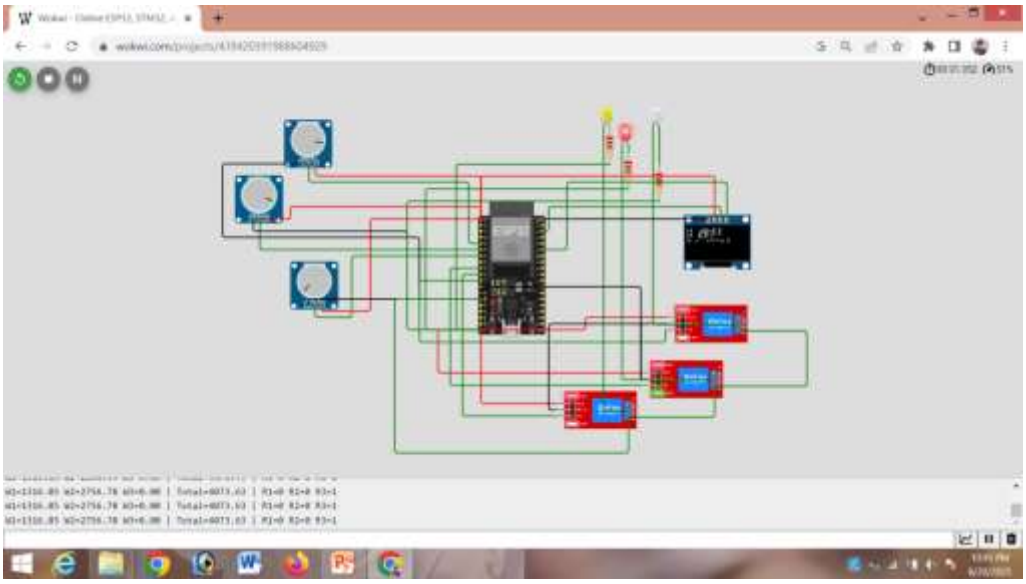
FLOWCHART

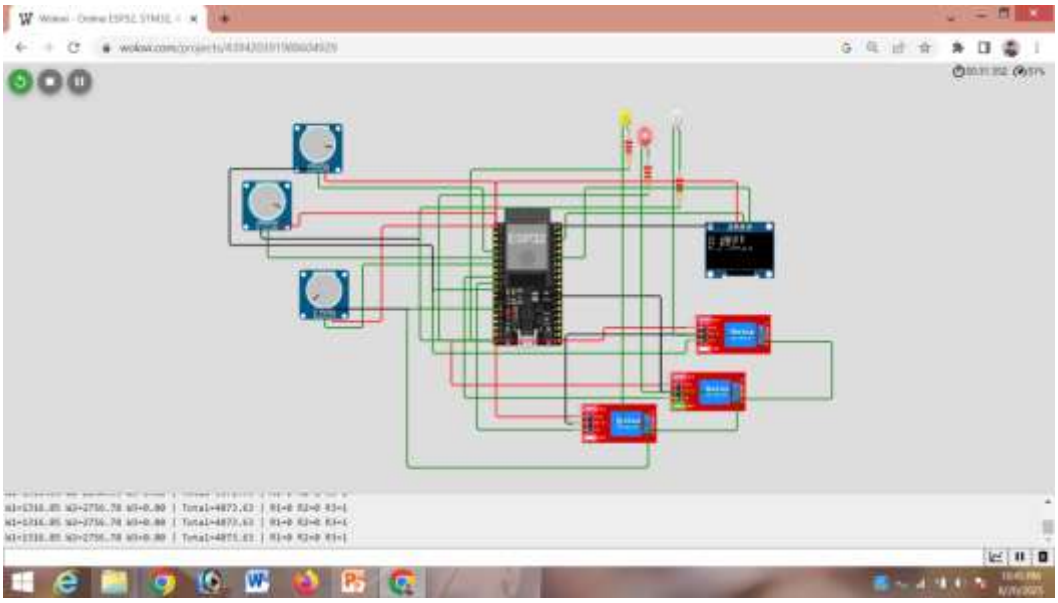

start
Power on the system(Li-ion) battery setup
Initialize ESP32 and sensors
Read voltage and current values from sensors
Calculate power P=V X I
If Bt low
Disconnect loads
If power exceeds limit
Turn of non-essential loads via relay
Continue normal position
Update real time data to mobile app
Display status locally in leds
Repeat monitering continuously
stop

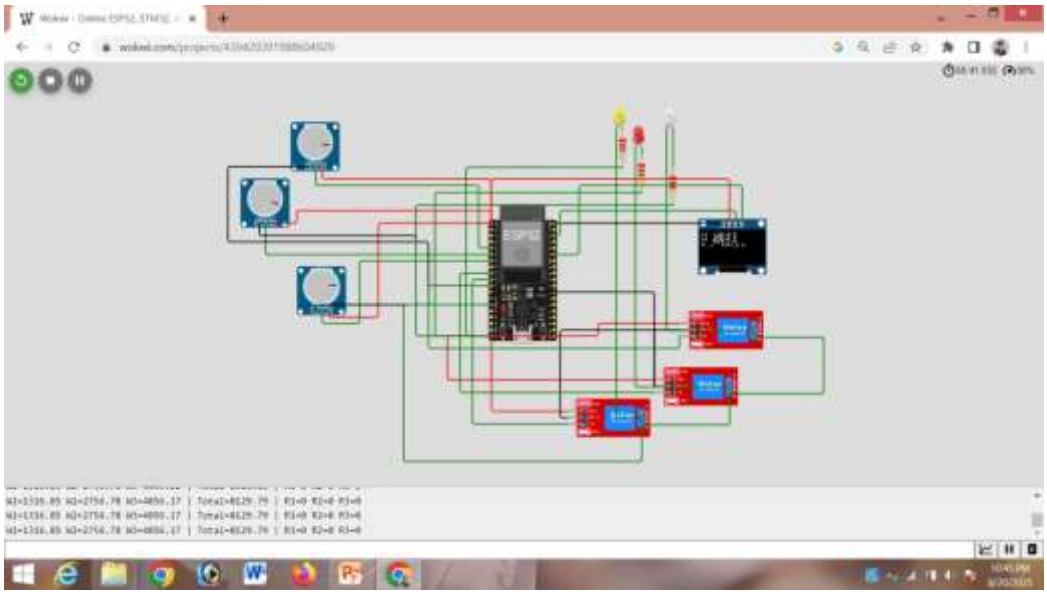
PROGRAM FLOW

- **Initialization: Sets up all hardware components, including the microcontroller, sensors, buzzer, and communication module (Wi-Fi/GSM).**
- **Continuous Monitoring: The core function of the system, involving constant reading, processing, and filtering of data from the Gas, Fire, and Smoke sensors.**
- **Threshold Detection: Uses pre-defined critical safety limits to determine if a hazard is present. This is the main decision-making point.**
- **Dual Alert System (Alarm Protocol):**
- **Local Alert: Immediate activation of the Buzzer/Siren and Visual Indicators (LEDs).**
- **Remote Notification: Using the communication module to send an alert (SMS/Cloud) to the user or authorities.**
- **Mitigation/Actuation (Optional): Includes automated response actions, such as activating an exhaust fan or shutting off a valve, to minimize danger.**
- **System Reset: Logic to confirm the environment is safe before deactivating the alarm and returning to the normal monitoring state.**

STIMULATION OUTPUT:







OBSERVATION:

During the testing phase of the smart load monitoring and control system, the sensors performed accurately in measuring real-time voltage, current, and power consumption of each connected load. The *ZMPT101B voltage sensor* provided a stable analog output corresponding to the AC voltage, while the *ACS712 current sensors* effectively measured the current drawn by individual loads such as the 9W bulb, fan, and LED strip. The data acquired was consistently processed by the *ESP32 microcontroller*, showing correct readings with minimal noise after calibration.When the total power consumption crossed the predefined threshold, the *ESP32 automatically activated the relay control logic, disconnecting non-essential loads in a sequential manner. This demonstrated the system's ability to prevent overload conditions and optimize battery usage efficiently.

Additionally, when the battery voltage dropped below a set limit, the controller successfully triggered alerts through the **buzzer and indicator LEDs*, confirming the reliability of the local alert system under low-power conditions.The system also showed effective communication with the *Blynk mobile application* via Wi-Fi. Real-time voltage, current, and power data were successfully transmitted and displayed on the mobile dashboard. Users could remotely monitor system parameters and load status, verifying that the IoT integration was functional and responsive.

REFERENCE

The development of this project was supported by various technical resources and documentation. The *ESP32 datasheet* from Espressif Systems was referred to for understanding the microcontroller’s Wi-Fi, ADC, and GPIO functionalities. The *ACS712 current sensor datasheet* by Allegro Microsystems provided detailed information on Hall-effect current sensing and output calibration. Similarly, the *ZMPT101B voltage sensor module reference* from ElectronicWings and RobotDyn helped in accurate AC voltage measurement and interfacing with the ESP32. The *Blynk IoT platform documentation* was used to establish wireless data transmission and mobile app control. Additionally, the *Arduino IDE documentation* from Arduino.cc was utilized for programming and simulation purposes. Finally, insights from research articles on *IoT-based smart energy monitoring systems* available on IEEE Xplore contributed to the design and analysis of efficient power monitoring and control strategies.